

2024 Ph H2 Q16

Section: Our Dynamic Universe

Topic: Hooke's Law & Elastic Energy

Summary of Question

Students investigate Hooke's Law using a spring and measure extension e (mm) for applied force F (N). They must: (a) draw F against e , (ii) determine spring constant k from the graph, (iii) suggest an improvement; (b) calculate elastic potential energy at 22 mm extension.

(a)(i) Graph of F against e

Plot F (N) on the y-axis against e (m) on the x-axis (convert mm $\rightarrow$ m).

Use a straight best-fit line through the data.

(a)(ii) Spring constant k from the graph

Hooke's Law: $F = k e$. The gradient of the straight line gives k (in N m^{-1}).

Using a least-squares best-fit to the provided data gives:

$k \approx 2.35 \times 10^2 \text{ N m}^{-1}$ (i.e. 235 N m^{-1}).

A gradient found directly from well-separated points on the line should be of order $2.4 \times 10^2 \text{ N m}^{-1}$.

(a)(iii) One improvement to the procedure

Examples (any one):

- Take more readings over a wider range/repeat and average to reduce random uncertainty.
- Use a pointer at the spring and take readings at eye-level to reduce parallax error.
- Ensure the spring is not over-stretched and allow it to come to rest before measuring.
- Measure the extension (change in length) rather than total length (zero the scale).

(b) Elastic potential energy at 22 mm

Relationship: $E_e = \frac{1}{2} k e^2$.

Using $e = 22 \text{ mm} = 0.022 \text{ m}$ and $k \approx 2.35 \times 10^2 \text{ N m}^{-1}$:

$$E_e = \frac{1}{2} \times (2.35 \times 10^2) \times (0.022)^2 \approx 5.69 \times 10^{-2} \text{ J} = 0.057 \text{ J}.$$

Final answer: $\approx 5.7 \times 10^{-2} \text{ J}$.

Revision Tips

Be consistent with units: convert mm $\rightarrow$ m before plotting or calculating.

If the best-fit line does not pass exactly through the origin, use its gradient rather than a single data point.

State k to sensible significant figures and include units (N m⁻¹).